Aniket Pradhan

Software Development Engineer

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Personal Profile

A nerd working at Adobe, as a Software Development Engineer. Currently working on the Streaming workflows in the Adobe Experience Platform, to segment billions of different ”profiles” within minutes. Zealous about Research and Development, with a few publications at major conferences around Optical Network Communication. Fascinated by Machine Learning and Data Engineering roles.

Work Experience

Adobe *San Jose, USA*

Software Development Engineer *June 2026 - Present*

- Contributed to core components of Nebula, a highly available, reactive metadata control plane that forms the backbone of Unified Profile Services (UPS), serving 50+ downstream consumers with millisecond-level read latency.
- Worked on near-real-time metadata propagation to consumers via the Hollow SDK, backed by Redis distributed caching and sandbox-level distributed locking to guarantee consistency under concurrent writes.
- Designed and developed edge-segments dataset provisioning within UPS, driving an event-driven flow (Kafka + Temporal workflows) to provision per-customer profile environments across Azure and AWS.
- **Technical Skills:** Java, Spring Boot, Project Reactor, PostgreSQL, Redis, Kafka, Temporal

Adobe *Noida, India*

Computer Scientist - 1 *July 2024 - May 2026*

- Led collaboration efforts to overhaul the internal Delta Lake, resulting in seamless data updates and optimized read performance.
- Contributed to an SDK that provided a structured framework for internal microservices to access the Internal Delta Lake Storage, boosting data accessibility and robustness for core Adobe Experience Platform features (Segmentation, Profile Export, Adobe Journey Optimizer).
- **Technical Skills:** Scala, Apache Spark, Delta-Lake, Databricks

Adobe *Noida, India*

Member of Technical Staff - 2 *Feb 2023 - June 2024*

- Drove cross-functional initiatives to streamline and improve the real-time segmentation process, ensuring smooth operation and continuous enhancement.
- Improved the stability and reliability of owned services by implementing key operational enhancements, significantly reducing points of failure.
- Developed and launched an internal dashboard providing microservice-level visibility into operational performance metrics; achieved rapid adoption across 50+ teams within the AEP and AJO units.
- **Technical Skills:** Scala, Apache Kafka, Python, Streamlit, Microsoft Azure, gRPC

Adobe *Noida, India*

Member of Technical Staff *Jul 2021 - Feb 2023*

- At Adobe, I am a part of the Unified Profile Services team, within the Adobe Experience Platform, where I work on the Streaming Segmentation and Edge Segmentation workflows.
- The work revolves around providing real-time segmentation for billions of user profiles around the world with minimal latency as low as in milliseconds.
- Dedicated efforts towards cost optimization for a sub-service, which resulted in a significant annual cost reduction exceeding $ 1 million in total.
- **Technical Skills:** Scala, Apache Kafka, Apache Spark, Microsoft Azure, gRPC

Adobe *Noida, India*

Product Intern *May 2020 - July 2020*

- As a Product Intern, I worked on a closed domain question and answering (QnA) framework, using the state-of-the-art methods of natural language processing.
- The QnA framework was designed to be flexible and modular, suitable for Adobe Experience Platform’s Documentation.
- **Technical Skills:** Python with Tensorflow and PyTorch XLA (for Google TPU), Git.

RealVol *New Delhi, India*

UI Designer Intern *May 2018 – July 2018*

- At RealVol I worked on the design and the UI for the product, RealVol
- **Technical Skills:** Unity VR, C#, Git.

OutWorx Solutions Pvt. Ltd. *Noida, India*

Android Developer Intern *June 2018 – July 2018*

- Created and implemented simple animations for Android.
- Implemented scalable widgets for Android Applications.
- **Technical Skills:** Android Studio, Java, Kotlin

Education

Indraprastha Institute of Information Technology - Delhi *Delhi, India*

B. Tech in Electronics and Communications Engineering *Aug 2017 - May 2021*

- **Courses:** Machine Learning, Deep Learning, Speech Recognition and Understanding, Game Design and Development, Software Development using Open Source, Digital Audio, Digital Image Recognition

Publications

- T. Ahmed, S. Rahman, A. Pradhan et al, "C to C+L Bands Upgrade with Resource Re-provisioning in Optical Backbone Networks", 2021 Optical Fiber Communications Conference and Exhibition (OFC), 2021, pp. 1-3.
- R K. Jana, A. Mitra, A. Pradhan et al, "When Multiband Elastic Optical Networks Becomes More Economical Than Multifiber Elastic Optical Networks?", European Conference on Optical Communication 2020, DOI: 10.1109/ECOC48923.2020.9333276
- A. Mitra, D. Semrau, N. Gahlawat, A. Pradhan et al, "Capacity Benefits of Operation Over C+L Band Elastic Optical Network in the Indian Network Scenario", IEEE ANTS 2019, DOI: 10.1109/ANTS47819.2019.9117992

University Projects

Efficient Filling of C+L Bands for an Optical Network *Delhi, India*

Indraprastha Insitute of Information Technology - Delhi and British Telecom *May 2019 - May 2020*

- Studied the non-linear effects faced by optical signals in a multi-hop path.
- Analyzed link margins in an optical network and studied the methods to reduce them in a backbone optical network.
- Compared the benefits offered by a multi-fiber optical network and a multi-band optical network.
- **Technical Skills:** Python, LaTeX.

Deep Neural Network Based Predictions of Protein Interactions using Primary Sequences *Delhi, India*

Indraprastha Institute of Information Technology - Delhi *Oct. 2018 - Feb. 2019*

- Created a tool in Python to automate data collection and organize the data in a pretty SQL table.
- Worked on and implemented Deep Structured Semantic Network (DSSM) in Python using Tensorflow
- Worked on an approach to simplify querying of similarity based protein sequences, using supervised methods
- **Technical Skills:** Python with Tensorflow, SQL.

Posters

- A. Sinha, A. Pradhan, Q. Fang et al, "Comp-NeuroFedora, a Free/Open Source operating system for computational neuroscience: download, install, research", Org. for Computational Neurosciences 2020